

Real-Time Fraud Detection System with Explainable AI and Live Dashboard

IEEE-CIS Fraud Detection — Capstone Project
AI and Data Analytics Internship | Week 4

LightGBM | XGBoost | Isolation Forest | SHAP | Optuna | Streamlit

1. Executive Summary

This project delivers an end-to-end machine learning pipeline for real-time credit card fraud detection on the IEEE-CIS dataset (590,000 transactions, 433 features, 3.5% fraud rate). The system addresses severe class imbalance, interpretability requirements, and operational deployment via a live dashboard.

- Best Model: LightGBM (Tuned) — ROC-AUC: 0.9734 | PR-AUC: 0.8241 | F1: 0.831
- SMOTE applied on training data only — no data leakage
- SHAP Explainable AI provides per-transaction plain-English justifications
- Risk segmentation into Critical / Suspicious / Clear tiers
- Estimated annual savings of \$302 million at 72% recall
- 3-page Streamlit dashboard deployed on Streamlit Community Cloud

2. Dataset — IEEE-CIS Fraud Detection

Property	Value
Source	Kaggle — IEEE-CIS Fraud Detection Competition
Files Used	train_transaction.csv + train_identity.csv
Total Transactions	590,540
Total Features	433 (after merge)
Fraud Rate	3.50% (20,663 fraud / 569,877 legit)
Join Key	TransactionID (left join)

3. Task Completion Summary

Task	Description	Status
Task 1	Data Loading, Merging and EDA	Complete
Task 2	Preprocessing, SMOTE, Feature Engineering	Complete
Task 3	Model Training, Comparison, Threshold Optimisation	Complete
Task 4	SHAP Explainable AI (Global + Per-Transaction)	Complete
Task 5	Risk Segmentation and Fraud Pattern Analysis	Complete
Task 6	Streamlit Multi-Page Dashboard (3 pages)	Complete
Task 7	5 Required Charts + Bonus Plotly Scatter	Complete
Task 8	Business Insights and Recommendations	Complete

4. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
LightGBM (Tuned)	0.9821	0.8743	0.7912	0.8308	0.9734	0.8241
XGBoost	0.9798	0.8512	0.7634	0.8050	0.9681	0.8017
Isolation Forest	0.9243	0.5821	0.6234	0.6021	0.8912	0.5634

LightGBM outperforms all models across every metric. Its gradient-boosting architecture handles the high-dimensional sparse feature space efficiently, while native leaf-wise tree growth captures complex fraud patterns that linear models cannot.

5. Feature Engineering

Feature	Formula / Logic	Fraud Signal
AmtToMeanRatio	TransactionAmt / mean(TransactionAmt)	Large amounts signal fraud
HourOfDay	(TransactionDT / 3600) mod 24	Fraud peaks 00:00-04:00
DeviceRisk	1 if DeviceType is null else 0	No fingerprint = higher risk

6. Risk Segmentation Results

Risk Tier	Probability	Transactions	Fraud Rate	Avg Amount	Action
Critical	>=0.75	1,321	83.1%	\$489.21	Block immediately
Suspicious	0.40-0.74	7,823	18.7%	\$213.67	Step-up auth
Clear	<0.40	55,412	1.2%	\$87.34	Allow

7. SHAP Explainability Insights

Top 3 SHAP-Identified Fraud Signals

- TransactionAmt / AmtToMeanRatio — Anomalous amounts are the single strongest predictor
- card1 / card2 — Specific BIN ranges exhibit elevated fraud rates
- addr1 / dist1 — Shipping vs billing address distance beyond 500km is a strong signal

Case	P(fraud)	Verdict	Key Feature	SHAP Value
Confirmed Fraud	0.9734	BLOCK	TransactionAmt	+0.412
Borderline	0.5023	FLAG	card1	+0.089

Case	P(fraud)	Verdict	Key Feature	SHAP Value
Legitimate	0.0187	ALLOW	AmtToMeanRatio	-0.287

8. Business Recommendations

Why PR-AUC Matters More Than Accuracy

With a 3.5% fraud rate, predicting all transactions as legitimate achieves 96.5% accuracy while catching zero fraud. PR-AUC directly measures minority-class detection without overwhelming analysts with false positives.

Fraud Prevention Policies

- Policy A — Velocity Block: Any card making 3+ transactions over \$500 within 10 minutes is auto-blocked. Targets card-testing and account takeover patterns.
- Policy B — Device Challenge: Transactions above \$200 with no device fingerprint require 2-FA SMS verification. Prevents anonymous high-value fraud.

Estimated Annual Impact

Metric	Value
Annual Transaction Volume	100,000,000
Fraud Rate	3.50%
Model Recall	72%
Transactions Blocked	2,520,000
Average Fraud Amount	\$120
Estimated Annual Savings	\$302,400,000

Model Limitations

- Dataset from 2017-2018 — evolving fraud patterns not fully captured
- SMOTE generates synthetic samples — may not generalise to novel attacks
- No temporal cross-validation — time-ordered data risks look-ahead bias